A Four-Direction Counterexample to First-Order Magnitude Sharpness

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Status. Candidate counterexample, likely valid, pending human review.

1 Source conjecture

The source is Mark W. Meckes, *Magnitude and Holmes–Thompson intrinsic volumes of convex bodies*, arXiv:2206.02600, published in the *Canadian Mathematical Bulletin* 66 (2023), 854–867 [1]. Conjecture 1.7 on page 4 (label `C:mu1-conj` in the source) states:

> Suppose that $E = (\mathbb{R}^n, \|\cdot\|)$ is a hypermetric normed space, and let $K \in \mathcal{K}^n$. Then
>
> $$\lim_{t\to 0^+} \frac{\mathrm{Mag}(tK, \|\cdot\|) - 1}{t} = \frac{1}{4}\mu_1^E(K).$$

We give a two-dimensional hypermetric norm and a full-dimensional square for which the limsup is strictly smaller than the conjectured coefficient.

2 Definitions and a spherical magnitude bound

For a compact positive definite metric space (X, d), the magnitude definition used in the source is

$$\mathrm{Mag}(X, d) = \sup_{\nu} \frac{\nu(X)^2}{\iint_{X\times X} e^{-d(x,y)}\, d\nu(x)\, d\nu(y)},$$

where the supremum is over finite signed Borel measures on X.

The following elementary lemma is the mechanism behind the counterexample.

Lemma 1 (Spherical Gaussian bound)**.** *Let X admit a map $\Phi : X \to H$ into a real Hilbert space such that*

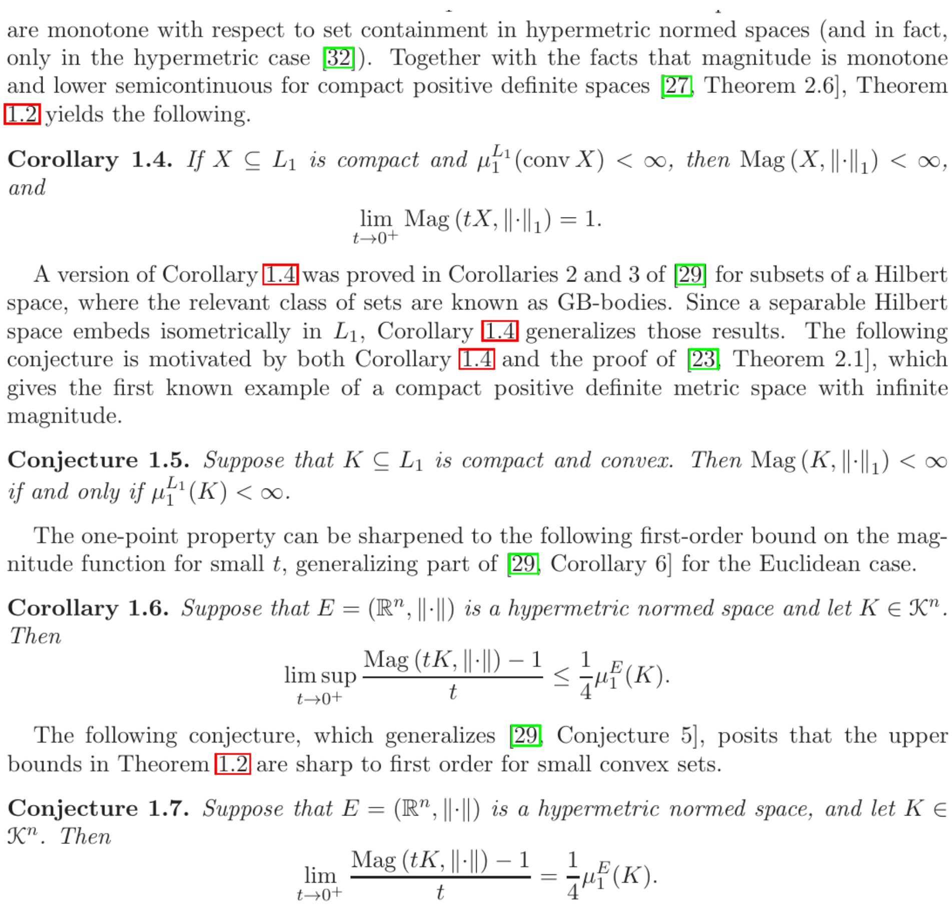
$$d(x, y) = \|\Phi(x) - \Phi(y)\|_H^2 .$$

If there are $p \in H$ and $R \geq 0$ such that $\|\Phi(x) - p\|_H = R$ for every $x \in X$, then, for every $t > 0$,

$$\mathrm{Mag}(X, td) \leq e^{2tR^2}.$$

Consequently,

$$\limsup_{t\to 0^+} \frac{\mathrm{Mag}(X, td) - 1}{t} \leq 2R^2.$$



are monotone with respect to set containment in hypermetric normed spaces (and in fact, only in the hypermetric case [32]). Together with the facts that magnitude is monotone and lower semicontinuous for compact positive definite spaces [27, Theorem 2.6], Theorem 1.2 yields the following.

Corollary 1.4. *If $X \subseteq L_1$ is compact and $\mu_1^{L_1}(\operatorname{conv} X) < \infty$, then $\operatorname{Mag}(X, \|\cdot\|_1) < \infty$, and*

$$\lim_{t \to 0^+} \operatorname{Mag}(tX, \|\cdot\|_1) = 1.$$

A version of Corollary 1.4 was proved in Corollaries 2 and 3 of [29] for subsets of a Hilbert space, where the relevant class of sets are known as GB-bodies. Since a separable Hilbert space embeds isometrically in L_1, Corollary 1.4 generalizes those results. The following conjecture is motivated by both Corollary 1.4 and the proof of [23, Theorem 2.1], which gives the first known example of a compact positive definite metric space with infinite magnitude.

Conjecture 1.5. *Suppose that $K \subseteq L_1$ is compact and convex. Then $\operatorname{Mag}(K, \|\cdot\|_1) < \infty$ if and only if $\mu_1^{L_1}(K) < \infty$.*

The one-point property can be sharpened to the following first-order bound on the magnitude function for small t, generalizing part of [29, Corollary 6] for the Euclidean case.

Corollary 1.6. *Suppose that $E = (\mathbb{R}^n, \|\cdot\|)$ is a hypermetric normed space and let $K \in \mathcal{K}^n$. Then*

$$\limsup_{t \to 0^+} \frac{\operatorname{Mag}(tK, \|\cdot\|) - 1}{t} \le \frac{1}{4}\mu_1^E(K).$$

The following conjecture, which generalizes [29, Conjecture 5], posits that the upper bounds in Theorem 1.2 are sharp to first order for small convex sets.

Conjecture 1.7. *Suppose that $E = (\mathbb{R}^n, \|\cdot\|)$ is a hypermetric normed space, and let $K \in \mathcal{K}^n$. Then*

$$\lim_{t \to 0^+} \frac{\operatorname{Mag}(tK, \|\cdot\|) - 1}{t} = \frac{1}{4}\mu_1^E(K).$$

Figure 1: Corollary 1.6 and Conjecture 1.7 in arXiv:2206.02600, p. 4.

Proof. Put $v_x = \Phi(x) - p$, so that $\|v_x\| = R$. For a finite signed Borel measure ν, uniform convergence of the exponential series on the radius-R sphere gives

$$\begin{aligned}
\iint e^{-td(x,y)}\, d\nu(x)d\nu(y) &= e^{-2tR^2} \iint e^{2t\langle v_x, v_y\rangle}\, d\nu(x)d\nu(y) \\
&= e^{-2tR^2} \sum_{m=0}^{\infty} \frac{(2t)^m}{m!} \left\| \int v_x^{\otimes m}\, d\nu(x) \right\|_{H^{\otimes m}}^2 \\
&\ge e^{-2tR^2} \nu(X)^2.
\end{aligned}$$

Every term is nonnegative. Substitution in the variational definition of magnitude yields the first assertion. The second follows from $(e^{2tR^2} - 1)/t \to 2R^2$. $\square$

3 Proof intuition

An L_1 metric is a squared Hilbert metric: each scalar coordinate a is sent to an oriented interval indicator η_a, for which $\|\eta_a - \eta_b\|_2^2 = |a - b|$. If the scalar coordinate ranges over $[-h, h]$, all these indicators lie on the sphere centered at the average of the two endpoint indicators, with squared radius $h/2$. Summing over the generating directions therefore gives the source paper's coefficient $\mu_1^E(K)/4$ through Lemma 1.

The important point is that this natural sphere need not be the smallest sphere available in the ambient Hilbert space. For the four directions used below, their rank-one quadratic tensors satisfy one nontrivial linear relation. That relation produces a Hilbert vector orthogonal to every embedded point but not orthogonal to the natural center. Removing the center's component in that direction strictly decreases the common radius. Lemma 1 then gives a strict first-order upper bound, which is incompatible with the conjectured equality.

4 The counterexample

Set

$$u_1 = (1,0), \qquad u_2 = (0,1), \qquad u_3 = \frac{(1,1)}{\sqrt{2}}, \qquad u_4 = \frac{(1,-1)}{\sqrt{2}},$$

and equip $\mathbb{R}^2$ with the norm

$$\|x\|_E = \sum_{j=1}^{4} |\langle x, u_j \rangle|. \tag{1}$$

Let $K = [-1,1]^2$.

Theorem 1. *The normed space $E = (\mathbb{R}^2, \|\cdot\|_E)$ is hypermetric and $K \in \mathcal{K}^2$, but*

$$\limsup_{t\to 0^+} \frac{\operatorname{Mag}(tK, \|\cdot\|_E) - 1}{t} \le 2 + 2\sqrt{2} - \frac{3}{2(1+2\sqrt{2})} < \frac{1}{4}\mu_1^E(K) = 2 + 2\sqrt{2}.$$

In particular, Conjecture 1.7 of the source paper is false.

Proof. Define the even nonnegative measure on S^1

$$\rho = \frac{1}{2}\sum_{j=1}^{4}(\delta_{u_j} + \delta_{-u_j}).$$

Then $\|x\|_E = \int_{S^1} |\langle x, u\rangle| \, d\rho(u)$, so Proposition 2.2 of the source paper shows that E is hypermetric. The set K is plainly a compact, full-dimensional convex body.

For $j = 1, \dots, 4$, put

$$h_j = \max_{x\in K} \langle x, u_j\rangle, \qquad (h_1, h_2, h_3, h_4) = (1, 1, \sqrt{2}, \sqrt{2}).$$

Let

$$H = \bigoplus_{j=1}^{4} L_2([-h_j, h_j]).$$

For $a \in [-h, h]$, define the oriented interval indicator

$$\eta_a(s) = \begin{cases} \mathbf{1}_{(0,a]}(s), & a \ge 0, \\ -\mathbf{1}_{(a,0]}(s), & a < 0. \end{cases}$$

It satisfies $\|\eta_a - \eta_b\|_{L_2}^2 = |a-b|$. Hence the map

$$\Phi(x) = \left(\eta_{\langle x,u_1\rangle}, \dots, \eta_{\langle x,u_4\rangle}\right)$$

obeys

$$\|\Phi(x) - \Phi(y)\|_H^2 = \sum_{j=1}^{4} |\langle x-y, u_j\rangle| = \|x-y\|_E. \tag{2}$$

For each j, let

$$c_j = \frac{1}{2}(\eta_{-h_j} + \eta_{h_j}), \qquad c = (c_1, c_2, c_3, c_4).$$

A direct one-dimensional calculation gives, for every $a \in [-h_j, h_j]$,

$$\|\eta_a - c_j\|_{L_2}^2 = \frac{h_j}{2}.$$

Thus every $\Phi(x)$, $x \in K$, lies on the sphere centered at c with squared radius

$$R_0^2 = \frac{1}{2}\sum_{j=1}^{4} h_j = 1 + \sqrt{2}. \tag{3}$$

We now strictly improve this center. Put

$$(\alpha_1, \alpha_2, \alpha_3, \alpha_4) = (-1, -1, 1, 1)$$

and define $g = (g_1, \dots, g_4) \in H$ by

$$g_j(s) = 2\alpha_j s \qquad (-h_j \le s \le h_j).$$

The four directions satisfy

$$u_3u_3^T + u_4u_4^T = u_1u_1^T + u_2u_2^T = I_2. \tag{4}$$

Since $\langle g_j, \eta_a\rangle_{L_2} = \alpha_j a^2$, equation (4) yields, for every $x \in K$,

$$\langle g, \Phi(x)\rangle_H = \sum_{j=1}^{4} \alpha_j \langle x, u_j\rangle^2 = 0. \tag{5}$$

On the other hand,

$$\langle g, c\rangle_H = \sum_{j=1}^{4} \alpha_j h_j^2 = 2, \tag{6}$$

$$\|g\|_H^2 = \frac{8}{3}\sum_{j=1}^{4} h_j^3 = \frac{16}{3}(1 + 2\sqrt{2}). \tag{7}$$

Let

$$q = \frac{\langle c, g\rangle}{\|g\|^2} g, \qquad p = c - q.$$

Then $p \perp g$, while $\Phi(x) \perp g$ by (5). Since $c = p + q$, the two summands in $\Phi(x) - c = (\Phi(x) - p) - q$ are orthogonal. Therefore all $\Phi(x)$ lie on the sphere centered at p whose squared radius is

$$R^2 = R_0^2 - \|q\|^2 = 1 + \sqrt{2} - \frac{3}{4(1 + 2\sqrt{2})}. \tag{8}$$

Combining (2), Lemma 1, and (8) gives

$$\limsup_{t\to 0^+} \frac{\operatorname{Mag}(tK, \|\cdot\|_E) - 1}{t} \le 2R^2 = 2 + 2\sqrt{2} - \frac{3}{2(1+2\sqrt{2})}. \tag{9}$$

It remains only to compare the normalization with the conjecture. For $m = 1$, Proposition 2.3 of the source paper has $c_1 = 2$, and hence

$$\mu_1^E(K) = 2\int_{S^1} \operatorname{vol}_1(A_u K)\, d\rho(u) = 2\sum_{j=1}^{4}(2h_j) = 8 + 8\sqrt{2}.$$

Consequently,

$$\frac{1}{4}\mu_1^E(K) = 2 + 2\sqrt{2}.$$

The right side of (9) is smaller by the positive amount

$$\frac{3}{2(1+2\sqrt{2})}.$$

This rules out the equality asserted by Conjecture 1.7. □

Remark 1. *The proof does not need the existence or value of the actual one-sided derivative. A strict limsup upper bound below the conjectured number already disproves the conjecture. It also does not use numerical approximation: all constants and the sphere reduction are exact.*

5 Verification notes

The companion script `code/verify_constants.py` uses exact symbolic arithmetic to check the rank-one tensor relation, the values of $\langle c, g\rangle$ and $\|g\|^2$, the radius drop, the $c_1 = 2$ normalization calculation, and the final strict gap. It was run with

```
conda run --no-capture-output -n sandbox python code/verify_constants.py
```

from the packet directory. The script is a sign and normalization regression check; Lemma 1 and Theorem 1 are the proof.

For human review, the most important points to inspect are:

- the use of finite signed measures in the tensor-series proof of Lemma 1;
- the orthogonal radius reduction in equations (5)–(8);
- the factor $c_1 = 2$ in the source paper's Proposition 2.3.

6 Novelty check and limitations

The run's lightweight indexes were searched first for arXiv:2206.02600 and the exact paper title. Additional searches used the terms Holmes–Thompson, first-order magnitude, and hypermetric magnitude. No prior registry, solution, attempt, or proof-gap entry for this conjecture was found.

A bounded web and arXiv search on 21 July 2026 used the exact title, arXiv identifier, Conjecture 1.7, sharp to first order, and hypermetric. It recovered the source paper and its 2019 Euclidean predecessor, but no later proof, counterexample, or matching four-direction construction. This is not an exhaustive MathSciNet or zbMATH citation search. Accordingly, the packet claims a likely valid mathematical counterexample but only moderate novelty confidence; an elementary construction of this kind could be unpublished folklore.

The result addresses only finite-dimensional Conjecture 1.7. It says nothing about Conjecture 1.5 in the source paper concerning finiteness of magnitude for compact convex subsets of infinite-dimensional L_1.

7 Human-review recommendation

Recommended verdict: **likely valid counterexample**. If the three checks listed above pass, the strict inequality directly contradicts the exact conclusion of Conjecture 1.7 while satisfying all its hypotheses.

References

[1] Mark W. Meckes, *Magnitude and Holmes–Thompson intrinsic volumes of convex bodies*, Canadian Mathematical Bulletin **66** (2023), no. 3, 854–867; arXiv:2206.02600; DOI: 10.4153/S0008439522000728.